

Section 9.1 - Evaluation Context

Evaluation context in DAX refers to the environment in which a DAX expression is evaluated. It determines the values that are considered during the calculation. There are two main types of evaluation contexts: row context and filter context. Row context is the context of the current row being processed, while filter context is the set of filters applied to the data model.

Row context in DAX is the context of the current row being processed. It allows DAX expressions to access the values of columns in the current row. For example, when creating a calculated column, DAX evaluates the expression for each row in the table, using the values of the columns in that row.

Filter context in DAX is the set of filters applied to the data model. It determines which rows are included in the calculation. Filter context can be created by slicers, filters, or DAX functions like CALCULATE. It allows DAX expressions to consider only the rows that meet certain criteria, making it possible to perform calculations on subsets of data.

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DAX evaluates all formulas within a respective context. Even though the formula is the same, the result is different because DAX executes the same code against different subsets of data.

Every cell of the report has a different filter context. You should consider that every cell has a different evaluation – as if it were a different query, independent from the other cells in the same report.

SQLBI mantra – The Row Context iterates, it doesn't filter. The Filter Context filters, it doesn't iterate.

Section 9.2 - Row Context

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The row context exists when we create a calculated column or when we are computing an expression inside an iteration. ***There is no other way of creating a row context.***

Since we are looking at a single row with a single value no aggregation is needed. The SUM of a single number equals the AVERAGE, MIN, MAX, MEDIAN of a single number.

9.2.1 – In Table view select the 'Product' table. Create a new column with BrandName and ClassName concatenated together with a space.

Brand+Class = 'Product'[BrandName] & " " & 'Product'[ClassName]

Notice it went row by row, looking at the current row and joined the two values together.

9.2.2 - What if we want profit per product item? We can create a calculated column with the following in the 'Product' table.

Profit = 'Product'[UnitPrice]-'Product'[UnitCost]

Notice we are only referencing the column names and Row Context is using the values in the current row of the two respective columns.

9.2.3 - You can even create pricing bands with SWITCH. Create a new Calculated column with the following formula in the 'Product' table.

```
Price Band =  
SWITCH (  
    TRUE (),  
    'Product'[UnitPrice] <= 100, "Low",  
    'Product'[UnitPrice] <= 1000, "Mid",  
    "High"  
)
```

9.2.4 - What happens if we use an aggregation in row context? Create a Calculated column with the following formula.

Profit with SUM = SUM('Product'[UnitPrice])-SUM('Product'[UnitCost])

We get the same value in every cell as there is no filter context

Section 9.3 – Row Context between tables

How do we use row context between tables? What if we want to see the sales amount for each product?

9.3.1 – Create a new calculated column with the following

Product Sales amount = `SUM(Sales[SalesAmount])`

We see the same result in each row. This represents the total sales of all products. Why? No filter context was passed to the 'Sales' table from the 'Product' table.

If we want to pass a filter context through a relationship we use `RELATED` or `RELATEDTABLE`

Section 9.4 – `RELATED` and `RELATEDTABLE`

`RELATED` and `RELATEDTABLE` are DAX functions used to work with row context across related tables. `RELATED` retrieves a value from a related table for the current row, while `RELATEDTABLE` returns a table of rows related to the current row. These functions help pass filter context through relationships, enabling more accurate calculations such as aggregating data for each product rather than for all rows combined.

9.4.1 – Let's retry the formula we used in 9.3.1 using `RELATEDTABLE`. Create a new column with the following

Product Sales amount 2= `SUMX(RELATEDTABLE(sales),Sales[SalesAmount])`

9.4.2 - We'll see shortly that this is what `calculate` does – passes a filter context using `RELATEDTABLE`. Enter the following as a new column

Product Sales amount 3= `CALCULATE(SUM(Sales[SalesAmount]))`

Notice it provides the same result as `RELATEDTABLE`

Section 9.5 – Nesting Row Context

9.5.1 - You can nest row context inside of another row context. In this example we use variables. Create a new column with the following

```
UnitPrice Rank =  
var currentunitprice = 'Product'[UnitPrice]  
var calculaterank = COUNTROWS(FILTER('Product', 'Product'[UnitPrice]>currentunitprice)) +1  
RETURN  
calculaterank
```

Notice this shows the top priced products as number 1 (4 of them) so the second highest priced products are 5th.

9.5.2 - What if we wanted to skip ties? (Show ranking as 1,2,3)

```
UnitPrice Rank Dense =  
var currentunitprice = 'Product'[UnitPrice]  
var calculaterank =  
COUNTROWS(FILTER(VALUES('Product'[UnitPrice]), 'Product'[UnitPrice]>currentunitprice)) +1  
RETURN  
calculaterank
```

By using VALUES this creates a new one column table of the unique UnitPrice amounts, then iterates through this new table checking the related unit price

Section 9.6 - Filter Context

Filter context in DAX refers to the set of filters that determine which data is included in a calculation. It is created automatically by visuals in tools like Power BI, but can also be modified using functions such as CALCULATE, FILTER, and ALL. By altering the filter context, you control what subset of data your formulas and aggregations operate on, allowing for flexible and dynamic calculations tailored to your analytical needs.

Overview of what is happening in a visual like below

ProductCategory	SalesAmount
Audio	\$151,614,364.31
Cameras and camcorders	\$2,562,023,774.06
Cell phones	\$892,233,264.30
Computers	\$3,209,427,141.42
Music, Movies and Audio Books	\$165,804,705.98
TV and Video	\$1,360,121,114.76
Total	\$8,341,224,364.83

This visual is using 7 different filter contexts (or versions of the fact table).

Section 9.7 - Calculate

Is there a way to alter what table is looking at to aggregate values? In other words is there a way to alter Filter Context? Calculate is the function that allows to modify filter context.

[CALCULATE - DAX Guide](#)

9.7.1 - Let's create a simple table visual using 'ProductCategory'[ProductCategory] and 'Sales'[SalesAmount]

ProductCategory	SalesAmount
Audio	\$151,614,364.31
Cameras and camcorders	\$2,562,023,774.06
Cell phones	\$892,233,264.30
Computers	\$3,209,427,141.42
Music, Movies and Audio Books	\$165,804,705.98
TV and Video	\$1,360,121,114.76
Total	\$8,341,224,364.83

What if we want to see the percentage of each product category of the total?

We need to alter the filter context, or the table that the formula looks at to aggregate the values

9.7.2 – Create a measure with the following formula

Intro to Calculate =

VAR SalesAmount = **SUM**(Sales[SalesAmount])

RETURN

SalesAmount

9.7.3 - Bring it into the visual like so

ProductCategory	SalesAmount	Intro to Calculate
Audio	\$151,614,364.31	\$151,614,364.31
Cameras and camcorders	\$2,562,023,774.06	\$2,562,023,774.06
Cell phones	\$892,233,264.30	\$892,233,264.30
Computers	\$3,209,427,141.42	\$3,209,427,141.42
Music, Movies and Audio Books	\$165,804,705.98	\$165,804,705.98
TV and Video	\$1,360,121,114.76	\$1,360,121,114.76
Total	\$8,341,224,364.83	\$8,341,224,364.83

The values match since the formula is performing the same aggregation on the same filter context.

To calculate the percentage of each productCategory of the total, we need a way to look at two different tables that we can aggregate. One specifically for the respective product category and one for the total.

In other words we need to alter the filter context.

9.7.4 - Does the filter context need to grow or shrink to get the total value?

9.7.5 - For this example the table needs to grow. This is accomplished by using Calculate with ALL.

Update the formula with:

Intro to Calculate =

`VAR SalesAmount = SUM(Sales[SalesAmount])`

`VAR FullSales = CALCULATE(SUM(Sales[SalesAmount]),ALL(ProductCategory[ProductCategory]))`

`RETURN`

`FullSales`

We now see the total on each row, as the filter context on each row is now the same

ProductCategory	SalesAmount	Intro to Calculate
Audio	\$151,614,364.31	\$8,341,224,364.83
Cameras and camcorders	\$2,562,023,774.06	\$8,341,224,364.83
Cell phones	\$892,233,264.30	\$8,341,224,364.83
Computers	\$3,209,427,141.42	\$8,341,224,364.83
Games and Toys		\$8,341,224,364.83
Home Appliances		\$8,341,224,364.83
Music, Movies and Audio Books	\$165,804,705.98	\$8,341,224,364.83
TV and Video	\$1,360,121,114.76	\$8,341,224,364.83
Total	\$8,341,224,364.83	\$8,341,224,364.83

9.7.6 - Update the formula with the final division to get the percentage

Intro to Calculate =

VAR SalesAmount = SUM(Sales[SalesAmount])

VAR FullSales = CALCULATE(SUM(Sales[SalesAmount]),ALL(ProductCategory[ProductCategory]))

VAR result = DIVIDE(SalesAmount,FullSales)

RETURN

result

ProductCategory	SalesAmount	Intro to Calculate
Audio	\$151,614,364.31	1.82%
Cameras and camcorders	\$2,562,023,774.06	30.72%
Cell phones	\$892,233,264.30	10.70%
Computers	\$3,209,427,141.42	38.48%
Music, Movies and Audio Books	\$165,804,705.98	1.99%
TV and Video	\$1,360,121,114.76	16.31%
Total	\$8,341,224,364.83	100.00%

9.7.7 – In the above example we altered the filter context by removing a filter from the filter context. In some cases we want to add a filter to the existing filter context. For example, instead of calculating the sales for all Brand Names, calculate the sales for just Contoso. Create a measure with the following:

Contoso Sales = CALCULATE(SUM(Sales[SalesAmount]),'Product'[BrandName]="Contoso")

It is important to identify what the new filter context that is being created is. The visual should look similar to this

ProductCategory	SalesAmount	Intro to Calculate	Contoso Sales
Audio	\$151,614,364.31	1.82%	\$65,190,616.40
Cameras and camcorders	\$2,562,023,774.06	30.72%	\$321,957,389.18
Cell phones	\$892,233,264.30	10.70%	\$218,707,856.34
Computers	\$3,209,427,141.42	38.48%	\$428,108,048.86
Music, Movies and Audio Books	\$165,804,705.98	1.99%	\$82,731,157.81
TV and Video	\$1,360,121,114.76	16.31%	\$381,225,699.77
Total	\$8,341,224,364.83	100.00%	\$1,497,920,768.36

9.7.8 – Create the following measure to see what percent Contoso sales is of total sales?

Contoso Sales Pct = `DIVIDE([Contoso Sales],SUM(Sales[SalesAmount]))`

ProductCategory	SalesAmount	Intro to Calculate	Contoso Sales	Contoso Sales Pct
Audio	\$151,614,364.31	1.82%	\$65,190,616.40	43.00%
Cameras and camcorders	\$2,562,023,774.06	30.72%	\$321,957,389.18	12.57%
Cell phones	\$892,233,264.30	10.70%	\$218,707,856.34	24.51%
Computers	\$3,209,427,141.42	38.48%	\$428,108,048.86	13.34%
Music, Movies and Audio Books	\$165,804,705.98	1.99%	\$82,731,157.81	49.90%
TV and Video	\$1,360,121,114.76	16.31%	\$381,225,699.77	28.03%
Total	\$8,341,224,364.83	100.00%	\$1,497,920,768.36	17.96%

It's important to be able to identify the filter contexts being used in this visual.

ProductCategory	SalesAmount	Intro to Calculate	Contoso Sales	Contoso Sales Pct
Audio	\$151,614,364.31	1.82%	\$65,190,616.40	43.00%
Cameras and camcorders	\$2,562,023,774.06	30.72%	\$321,957,389.18	12.57%
Cell phones	\$892,233,264.30	10.70%	\$218,707,856.34	24.51%
Computers	\$3,209,427,141.42	38.48%	\$428,108,048.86	13.34%
Music, Movies and Audio Books	\$165,804,705.98	1.99%	\$82,731,157.81	49.90%
TV and Video	\$1,360,121,114.76	16.31%	\$381,225,699.77	28.03%
Total	\$8,341,224,364.83	100.00%	\$1,497,920,768.36	17.96%

The diagram illustrates the filter contexts for the 'Contoso Sales Pct' measure. It uses colored boxes and arrows to show the data sources for each cell in the table:

- Audio SalesAmount (\$151,614,364.31):** Filtered by ProductCategory (Audio).
- Audio Intro to Calculate (1.82%):** Filtered by ProductCategory (Audio).
- Audio Contoso Sales (\$65,190,616.40):** Filtered by ProductCategory (Audio).
- Audio Contoso Sales Pct (43.00%):** Filtered by ProductCategory (Audio).
- Total SalesAmount (\$8,341,224,364.83):** Filtered by ProductCategory (Total).
- Total Intro to Calculate (100.00%):** Filtered by ProductCategory (Total).
- Total Contoso Sales (\$1,497,920,768.36):** Filtered by ProductCategory (Total).
- Total Contoso Sales Pct (17.96%):** Filtered by ProductCategory (Total).

Each color represents the filter context or table that the cell is using to aggregate values out of.

9.7.9 - What happens if we alter the visual to filter by BrandName and use the Contoso Sales Measure like below?

BrandName	SalesAmount	Contoso Sales
A. Datum	\$619,803,753.56	\$1,497,920,768.36
Adventure Works	\$1,089,734,248.74	\$1,497,920,768.36
Contoso	\$1,497,920,768.36	\$1,497,920,768.36
Fabrikam	\$1,874,455,854.35	\$1,497,920,768.36
Litware	\$327,894,810.44	\$1,497,920,768.36
Northwind Traders	\$20,163,003.97	\$1,497,920,768.36
Proseware	\$954,540,220.53	\$1,497,920,768.36
Southridge Video	\$471,653,445.03	\$1,497,920,768.36
The Phone Company	\$673,525,407.96	\$1,497,920,768.36
Wide World Importers	\$811,532,851.91	\$1,497,920,768.36
Total	\$8,341,224,364.83	\$1,497,920,768.36

Calculate has a built in filter condition that uses Filter with ALL, that then overwrites the original filter context. This is the same as

Contoso Sales =

```
CALCULATE(SUM(Sales[SalesAmount]),FILTER(ALL('Product'[BrandName]),'Product'[BrandName]
="Contoso"))
```

Which is also the same as

Contoso Sales =

```
VAR Newtable = FILTER(ALL('Product'[BrandName]),'Product'[BrandName] = "Contoso")
VAR Result = CALCULATE(SUM(Sales[SalesAmount]),Newtable)
RETURN
Result
```

9.7.10 - What if we remove the ALL in the formula?

Contoso Sales =

```
VAR Newtable = FILTER(VALUES('Product'[BrandName]),'Product'[BrandName] = "Contoso")
VAR Result = CALCULATE(SUM(Sales[SalesAmount]),Newtable)
RETURN
Result
```

We now follow filter context in the visual and return the following:

BrandName	SalesAmount	Contoso Sales
A. Datum	\$619,803,753.56	
Adventure Works	\$1,089,734,248.74	
Contoso	\$1,497,920,768.36	\$1,497,920,768.3562
Fabrikam	\$1,874,455,854.35	
Litware	\$327,894,810.44	
Northwind Traders	\$20,163,003.97	
Proseware	\$954,540,220.53	
Southridge Video	\$471,653,445.03	
The Phone Company	\$673,525,407.96	
Wide World Importers	\$811,532,851.91	
Total	\$8,341,224,364.83	\$1,497,920,768.3562

Which is the same as

Contoso Sales =

```
CALCULATE(SUM(Sales[SalesAmount]),FILTER('Product','Product'[BrandName]="Contoso"))
```

9.7.11 - Quick introduction to KEEPFILTERS – this removes the filter in the predicate (BrandName=Contoso) and then follows the filter context of the visual. So it alters the table being aggregated, and then adds back the original filter context.

Contoso Sales =

`CALCULATE(SUM(Sales[SalesAmount]),KEEPFILTERS('Product'[BrandName]="Contoso"))`

BrandName	SalesAmount	Contoso Sales
A. Datum	\$619,803,753.56	
Adventure Works	\$1,089,734,248.74	
Contoso	\$1,497,920,768.36	\$1,497,920,768.3562
Fabrikam	\$1,874,455,854.35	
Litware	\$327,894,810.44	
Northwind Traders	\$20,163,003.97	
Proseware	\$954,540,220.53	
Southridge Video	\$471,653,445.03	
The Phone Company	\$673,525,407.96	
Wide World Importers	\$811,532,851.91	
Total	\$8,341,224,364.83	\$1,497,920,768.3562

9.7.12 - This is more clear if we look at two BrandNames in the filter context.

Contoso Sales =

`CALCULATE(SUM(Sales[SalesAmount]),'Product'[BrandName] IN { "Contoso","Litware"})`

BrandName	SalesAmount	Contoso Sales
A. Datum	\$619,803,753.56	\$1,825,815,578.7952
Adventure Works	\$1,089,734,248.74	\$1,825,815,578.7952
Contoso	\$1,497,920,768.36	\$1,825,815,578.7952
Fabrikam	\$1,874,455,854.35	\$1,825,815,578.7952
Litware	\$327,894,810.44	\$1,825,815,578.7952
Northwind Traders	\$20,163,003.97	\$1,825,815,578.7952
Proseware	\$954,540,220.53	\$1,825,815,578.7952
Southridge Video	\$471,653,445.03	\$1,825,815,578.7952
The Phone Company	\$673,525,407.96	\$1,825,815,578.7952
Wide World Importers	\$811,532,851.91	\$1,825,815,578.7952
Total	\$8,341,224,364.83	\$1,825,815,578.7952

We now get the value of Contoso and Litware combined on each row.

9.7.13 - KEEPFILTERS now looks at a table of just Contoso and Litware and then honors the filter context.

Contoso Sales =

`CALCULATE(SUM(Sales[SalesAmount]),KEEPFILTERS('Product'[BrandName] IN {
"Contoso","Litware"}))`

BrandName	SalesAmount	Contoso Sales
A. Datum	\$619,803,753.56	
Adventure Works	\$1,089,734,248.74	
Contoso	\$1,497,920,768.36	\$1,497,920,768.3562
Fabrikam	\$1,874,455,854.35	
Litware	\$327,894,810.44	\$327,894,810.439
Northwind Traders	\$20,163,003.97	
Proseware	\$954,540,220.53	
Southridge Video	\$471,653,445.03	
The Phone Company	\$673,525,407.96	
Wide World Importers	\$811,532,851.91	
Total	\$8,341,224,364.83	\$1,825,815,578.7952

Section 9.8 - Context Transition

Context transition in DAX refers to the process by which a row context, typically created when iterating over a table (such as in a calculated column), is converted into a filter context that affects calculations on related tables. This transition is essential when calculating aggregates like sums or averages per row, ensuring that the calculation only considers data relevant to the current row. Functions like CALCULATE or iterators like SUMX with RELATEDTABLE invoke context transition, allowing DAX formulas to dynamically apply filters based on the current row's values and produce accurate results in measures and calculated columns.

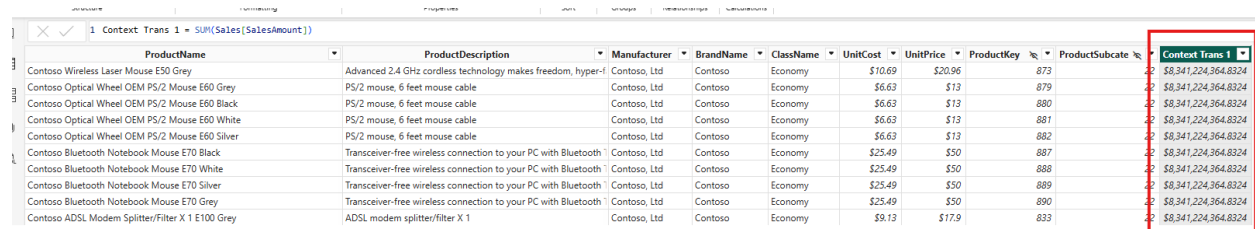
Converts ROW context into FILTER context

9.8.1 - In table view let's look at the Product table again. We have a list of the products and want to see the total sales amount for each product.

If we create a new calculated column that will invoke row context. Create a Calculated Column with the following formula:

Context Trans 1 = SUM(Sales[SalesAmount])

9.8.2 – Notice we get the full sales amount from the Sales Table. Why? Row context iterates, it doesn't filter, so there was no filter passed to the sales table.

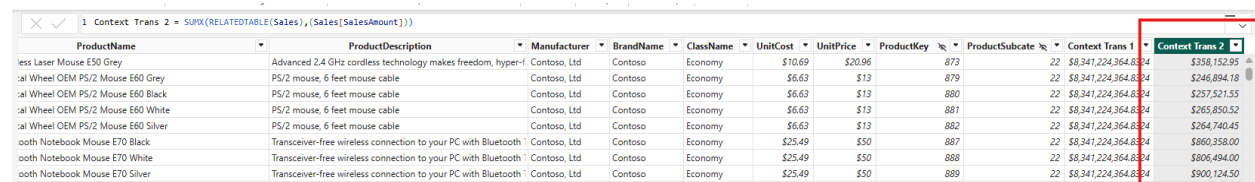


Product Name	Product Description	Manufacturer	Brand Name	Class Name	Unit Cost	Unit Price	Product Key	Product Subcategory	Context Trans 1
Contoso Wireless Laser Mouse E50 Grey	Advanced 2.4 GHz cordless technology makes freedom, hyper-	Contoso, Ltd	Contoso	Economy	\$10.69	\$20.96	873		\$8,341,224,364.8324
Contoso Optical Wheel OEM PS/2 Mouse E60 Grey	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	879		\$8,341,224,364.8324
Contoso Optical Wheel OEM PS/2 Mouse E60 Black	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	880		\$8,341,224,364.8324
Contoso Optical Wheel OEM PS/2 Mouse E60 White	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	881		\$8,341,224,364.8324
Contoso Optical Wheel OEM PS/2 Mouse E60 Silver	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	882		\$8,341,224,364.8324
Contoso Bluetooth Notebook Mouse E70 Black	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	887		\$8,341,224,364.8324
Contoso Bluetooth Notebook Mouse E70 White	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	888		\$8,341,224,364.8324
Contoso Bluetooth Notebook Mouse E70 Silver	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	889		\$8,341,224,364.8324
Contoso Bluetooth Notebook Mouse E70 Grey	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	890		\$8,341,224,364.8324
Contoso ADSL Modem Splitter/Filter X 1 E100 Grey	ADSL modem splitter/filter X 1	Contoso, Ltd	Contoso	Economy	\$9.13	\$17.9	833		\$8,341,224,364.8324

9.8.3 - To pass a filter we would need to involve a function that can pass a filter through a relationship – RELATEDTABLE (Back in section 9.4. we used this). Create a new column with the following:

Context Trans 2 = SUMX(RELATEDTABLE(Sales),(Sales[SalesAmount]))

This creates context transition, or turns all values on the current row into a filter context that gets passed to the sales table. We change to SUMX as RELATEDTABLE needs a table argument, and SUM only provides a column argument.



Product Name	Product Description	Manufacturer	Brand Name	Class Name	Unit Cost	Unit Price	Product Key	Product Subcategory	Context Trans 1	Context Trans 2
less Laser Mouse E50 Grey	Advanced 2.4 GHz cordless technology makes freedom, hyper-	Contoso, Ltd	Contoso	Economy	\$10.69	\$20.96	873		22 \$8,341,224,364.8324	\$358,152.95
al Wheel OEM PS/2 Mouse E60 Grey	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	879		22 \$8,341,224,364.8324	\$246,894.18
al Wheel OEM PS/2 Mouse E60 Black	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	880		22 \$8,341,224,364.8324	\$257,521.55
al Wheel OEM PS/2 Mouse E60 White	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	881		22 \$8,341,224,364.8324	\$265,850.52
al Wheel OEM PS/2 Mouse E60 Silver	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	882		22 \$8,341,224,364.8324	\$264,740.45
ooth Notebook Mouse E70 Black	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	887		22 \$8,341,224,364.8324	\$860,358.00
ooth Notebook Mouse E70 White	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	888		22 \$8,341,224,364.8324	\$806,494.00
ooth Notebook Mouse E70 Silver	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	889		22 \$8,341,224,364.8324	\$900,124.50

9.8.4 - However, if we use Calculate we see that Calculate provides context transition.

Context Trans 3 = **CALCULATE**(SUM(Sales[SalesAmount]))

1 Context Trans 3 = CALCULATE(SUM(Sales[SalesAmount]))											
me	ProductDescription	Manufacturer	BrandName	ClassName	UnitCost	UnitPrice	ProductKey	ProductSubcate	Context Trans 1	Context Trans 2	Context Trans 3
	Advanced 2.4 GHz cordless technology makes freedom, hyper-f	Contoso, Ltd	Contoso	Economy	\$10.69	\$20.96	873	22	\$8,341,224,364.8324	\$358,152.95	\$358,152.95
e E60 Grey	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	879	22	\$8,341,224,364.8324	\$246,894.18	\$246,894.18
e E60 Black	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	880	22	\$8,341,224,364.8324	\$257,521.55	\$257,521.55
e E60 White	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	881	22	\$8,341,224,364.8324	\$265,850.52	\$265,850.52
e E60 Silver	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	882	22	\$8,341,224,364.8324	\$264,740.45	\$264,740.45
0 Black	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	887	22	\$8,341,224,364.8324	\$860,358.00	\$860,358.00
0 White	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	888	22	\$8,341,224,364.8324	\$806,494.00	\$806,494.00
0 Silver	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	889	22	\$8,341,224,364.8324	\$900,124.50	\$900,124.50

9.8.5 - Finally, if we create a measure to aggregate SalesAmount using the following formula

SalesAmountSUM = SUM(Sales[SalesAmount])

we see something interesting if we use the measure in the new column. Create a new column using the above measure.

Context Trans 4 = [SalesAmountSUM]

1 Context Trans 4 = {SalesAmountSUM}												
	ProductDescription	Manufacturer	BrandName	ClassName	UnitCost	UnitPrice	ProductKey	ProductSubcate	Context Trans 1	Context Trans 2	Context Trans 3	Context Trans 4
	Advanced 2.4 GHz cordless technology makes freedom, hyper-f	Contoso, Ltd	Contoso	Economy	\$10.69	\$20.96	873	22	\$8,341,224,364.8324	\$358,152.95	\$358,152.95	\$358,152.95
	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	879	22	\$8,341,224,364.8324	\$246,894.18	\$246,894.18	\$246,894.18
	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	880	22	\$8,341,224,364.8324	\$257,521.55	\$257,521.55	\$257,521.55
	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	881	22	\$8,341,224,364.8324	\$265,850.52	\$265,850.52	\$265,850.52
	PS/2 mouse, 6 feet mouse cable	Contoso, Ltd	Contoso	Economy	\$6.63	\$13	882	22	\$8,341,224,364.8324	\$264,740.45	\$264,740.45	\$264,740.45
	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	887	22	\$8,341,224,364.8324	\$860,358.00	\$860,358.00	\$860,358.00
	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	888	22	\$8,341,224,364.8324	\$806,494.00	\$806,494.00	\$806,494.00
	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	889	22	\$8,341,224,364.8324	\$900,124.50	\$900,124.50	\$900,124.50
	Transceiver-free wireless connection to your PC with Bluetooth	Contoso, Ltd	Contoso	Economy	\$25.49	\$50	890	22	\$8,341,224,364.8324	\$851,729.50	\$851,729.50	\$851,729.50
	ADSL modem splitter/filter X 1	Contoso, Ltd	Contoso	Economy	\$9.13	\$17.9	833	22	\$8,341,224,364.8324	\$163,425.57	\$163,425.57	\$163,425.57

When a measure is referenced inside another DAX formula, it comes with a hidden calculate, which brings with it context transition.

Section 9.10 - Time Intelligence

Standard time-related calculations – DAX Patterns

Mark the date table as a date table

Difference between Time Intelligence functions and Time Date functions – review DAX.guide

9.10.1 – Create a measure to Calculate last month Sales:

LM Sales = `CALCULATE(SUM(Sales[SalesAmount]),DATEADD('Calendar'[DateKey],-1,MONTH))`

9.10.2 – Create a measure to calculate the MoM growth

Calculate MoM Growth

LM Sales Growth =

`VAR currentmonth = [SalesAmount SUM]`

`VAR lastmonth = [LM Sales]`

`VAR delta = currentmonth-lastmonth`

`RETURN`

`delta`

9.10.3 – Create a measure to calculate the MoM growth as a percent

Calculate MoM Growth %

LM Sales Growth % = `DIVIDE([LM Sales Growth],[LM Sales])`

9.10.4 – Create a measure to calculate the Sales for the same month a year prior

SAMEPERIODELASTYEAR

LY Sales = `CALCULATE([SalesAmount SUM],SAMEPERIODELASTYEAR('Calendar'[DateKey]))`

9.10.5 - Create a measure to calculate the Sales year to date

TOTALYTD

YTD Sales = `TOTALYTD([SalesAmount SUM],'Calendar'[DateKey])`

9.10.6 - Create a measure to calculate the Sales year to date for the previous year

PY YTD

PY YTD Sales = `TOTALYTD([SalesAmount SUM],SAMEPERIODELASTYEAR('Calendar'[DateKey]))`

9.10.7 - Create a measure to calculate the Sales delta YTD for this year vs last year

YOY Delta

YoY Delta = [YTD Sales]-[PY YTD]

9.10.8 - Create a measure to calculate the Sales delta YTD for this year vs last year

YTD YOY GROWTH

YTD YoY Growth = DIVIDE([YoY Delta],[PY YTD Sales])